Lab 06: Datapath Components I

Started: Oct 16 at 10:44am

Quiz Instructions

Introduction

The design in this lab will demonstrate the ways in which Verilog encoding makes hardware design more efficient. In particular, you will design the Sign Extender and ALU units for your ARMv8 processor. You will also explore directed random testing methodologies.

Pre-requisite

For this lab you are expected to know Verilog programming, understand how to use iVerilog and GTKWave based upon the last lab and understand the ALU and Sign Extender from Appendix A.5 of the textbook and Prelab 06.

Note: the modules that you implement in this lab and several labs following will be used as building blocks for a complete single cycle processor. It is highly recommended that you implement and test your models properly.

Question 1 50 pts

Build and run the Sign Extender you designed in the prelab though the testbench you designed for it.

If any of the tests fail, debug them and fix your code.

Bring up GTKWave on the VCD dump file your testbench creates.

o for your TA. Your demo should include:

- You walking the TA through your sign extender and testbench Verilog code. You should be able to explain how both work.
- You showing the test playing out in the GTKWave window

In the space below cut and paste the output from your testbench.

Discuss in the space below, discuss any challenges you might have from your testing strategy on this design. Given it is not exhaustive, could a design bug make its way into the final product? What could you do to improve your testbench to make bugs less likely?

Question 2 50 pts

Build and run the ALU you designed in the prelab though the testbench you designed for it.

If any of the tests fail, debug them and fix your code.

Bring up GTKWave on the VCD dump file your testbench creates.

Demo for your TA. Your demo should include:

- You walking the TA through your sign extender and testbench Verilog code. You should be able to explain how both work.
  You showing the test playing out in the GTKWave window

In the space below cut and paste the output from your testbench.

Discuss in the space below, discuss any challenges you might have from your testing strategy on this design.

ARMv8 has unsigned immediate ALU operations. How can they be used with signed numbers without causing problems?

Question 3 50 pts

Build and run the NextPC unit you designed in the prelab though the testbench you designed for it.

If any of the tests fail, debug them and fix your code.

Bring up GTKWave on the VCD dump file your testbench creates.

Demo for your TA. Your demo should include:

- You walking the TA through your sign extender and testbench Verilog code. You should be able to explain how both work.
  You showing the test playing out in the GTKWave window

In the space below cut and paste the output from your testbench.

Discuss in the space below, discuss any challenges you might have from your testing strategy on this design.

Note that the CurrentPC and NextPC are both unsigned, while the SignExtImm64 is signed. Are there any potential issues with this arrangement? What happens in your code on under- or over-flow?

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